IOT BASED UNDERGROUND COAL MINING ROVER FOR TOXIC GAS DETECTION ENVIRONMENTAL MONITORING AND REAL TIME SURVEILLANCE

EEC1361 SOLAR ENERGY TECHNOLOGY

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in

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BONAFIDE CERTIFICATE

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VISION

To emerge as a leader among the top institutions in the field of technical education.

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- Produce smart technocrats with empirical knowledge who can surmount the global challenges.
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- **PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
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- **PSO3 – Electronics and Instrumentation Engineering:** Design, Develop and implement methods and concepts to facilitate solutions for electrical and electronics engineering-related real-world problems.

Abstract (keywords)	POs mapping
Underground Coal Mining, IoT-Based Rover, ESP32 Microcontroller, ESP32-CAM Surveillance, Toxic Gas Detection, MQ-9 and MQ-135 Gas Sensors, Environmental Monitoring, ThingSpeak and Blynk, Real-Time Hazard Detection, Mine Worker Safety.	PO1, PO2, PO3, PO5, PO6, PO7,PSO ,PSO 2,PSO 3

SDG MAPPING

SDG Goal		Remarks
SDG 3	Good Health and Well-being	The IoT-based coal mining rover supports worker health and safety through remote monitoring of gas levels, temperature, humidity, light and moisture. Its camera and cloud-based monitoring help operators observe potentially hazardous areas and identify abnormal conditions, reducing the need for immediate direct inspection.

SDG Goal		Remarks
SDG 8	Decent Work and Economic Growth	The mobile rover supports safer working practices by enabling remote inspection of underground mining areas. Its environmental sensors, live camera, local display and wireless monitoring provide useful information about potentially hazardous conditions while helping reduce workers' direct exposure during preliminary inspections.
SDG 9	Industry, Innovation and Infrastructure	The project promotes engineering innovation by integrating an ESP32 controller, environmental sensors, mobile robotics, camera surveillance and IoT communication. ThingSpeak supports remote data monitoring, while Blynk enables rover control, demonstrating a connected monitoring prototype for mining-safety applications.

IEEE PROFESSIONAL STANDARDS

IEEE Standards Number	IEEE Standards Name	Remarks
IEEE 2413-2019	IEEE Standard for an Architectural Framework for the Internet of Things (IoT)	Provides an architectural framework for IoT systems, relevant to the sensor, controller and cloud-monitoring structure of the rover.
IEEE 802.11	IEEE Standard for Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications	Defines the Wi-Fi communication used by the ESP32 for ThingSpeak data upload and Blynk remote control.

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ABSTRACT

Underground coal mines are hazardous environments where toxic gases, high temperature, excessive moisture and poor illumination create serious risks for workers. This project presents an IoT-Based Underground Coal Mining Rover that allows remote monitoring and surveillance of underground areas without sending people into dangerous locations. Conventional inspection with portable detectors and fixed monitoring points exposes workers to risk and cannot cover the whole mine, which makes a mobile, low-cost and remotely operated solution necessary.An ESP32 controller collects data from MQ-9 and MQ-135 gas sensors, a DHT11 temperature and humidity sensor, an LDR light sensor and a moisture sensor. An ESP32-CAM mounted on a servo motor that sweeps from 0° to 180° provides live visual surveillance. The sensor data is displayed on a 16 × 2 LCD and uploaded every 20 seconds to the ThingSpeak cloud as six fields, while the rover is driven forward, backward, left and right through an L293D motor driver using the Blynk application. A regulated supply provides 12 V for the motors and 5 V for the electronics, and a buzzer alerts the operator to hazardous conditions.The prototype is built from readily available components at an approximate cost of Rs. 3810. The hardware output and the ThingSpeak graphs show that the rover measures gas, temperature, humidity, light and moisture values, uploads them to the cloud and provides live video. The system supports early detection of hazardous conditions, reduces human exposure and enables a faster response to danger. Future work includes AI-based prediction, long-range communication, autonomous navigation and a safe flameproof enclosure.

Keywords : Underground Coal Mining, IoT-Based Rover, ESP32 Microcontroller, ESP32-CAM Surveillance, Toxic Gas Detection, MQ-9 and MQ-135 Gas Sensors, Environmental Monitoring, ThingSpeak and Blynk, Real-Time Hazard Detection, Mine Worker Safety.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	FULL DESCRIPTION
1	IoT	Internet of Things
2	ESP32-CAM	ESP32 Camera Module
3	ADC	Analog-to-Digital Converter
4	GPIO	General Purpose Input/Output
5	LCD	Liquid Crystal Display
6	LDR	Light Dependent Resistor
7	DHT11	Digital Humidity and Temperature Sensor
8	PWM	Pulse-Width Modulation
9	DC	Direct Current
10	AC	Alternating Current
11	RPM	Revolutions Per Minute
12	RH	Relative Humidity
13	CO	Carbon Monoxide
14	IDE	Integrated Development Environment
15	API	Application Programming Interface
16	WSN	Wireless Sensor Network
17	GSM	Global System for Mobile Communications
18	LoRa	Long Range
19	LiDAR	Light Detection and Ranging
20	SDG	Sustainable Development Goal
21	Wi-Fi	Wireless Fidelity
22	CH_4	Methane
23	CO_2	Carbon Dioxide
24	LPG	Liquefied Petroleum Gas

CHAPTER 1
INTRODUCTION

Coal continues to be one of the most important sources of energy for electricity generation and for industries such as steel, cement and fertiliser production. Underground coal mining, however, is carried out in confined, poorly ventilated and poorly lit tunnels where workers are exposed to toxic gases, high humidity, water seepage and the constant risk of fire or explosion. Ensuring the safety of the people who work in such places remains one of the most serious engineering challenges of the mining industry.The proposed project, an IoT-Based Underground Coal Mining Rover for Toxic Gas Detection, Environmental Monitoring and Real-Time Surveillance, addresses this challenge by sending a small remotely operated rover into the mine instead of a human inspector. The rover is built around an ESP32 controller. It carries MQ-9 and MQ-135 gas sensors, a DHT11 temperature and humidity sensor, an LDR light sensor and a moisture sensor, together with a servo-mounted ESP32-CAM for live video. The measured values are shown on a local LCD and are uploaded to the ThingSpeak cloud, while the rover is driven from a mobile phone through the Blynk application. Overall, the system aims to detect hazardous conditions early, reduce the exposure of workers, and support a faster and better-informed response to danger.

1.1 BACKGROUND OF THE WORK

Underground coal seams release gases that are harmful in different ways. Methane, the main constituent of firedamp, is colourless and odourless and can form an explosive mixture with air when its concentration lies roughly between 5% and 15% by volume. Carbon monoxide is produced by incomplete combustion, mine fires and blasting; it is toxic even at low concentrations because it prevents the blood from carrying oxygen. Other gases, smoke and dust further reduce air quality. Because most of these gases cannot be seen or smelled, workers depend entirely on instruments to know whether the air is safe.Besides gases, the physical environment of a mine affects both safety and equipment. High temperature and humidity cause heat stress and fatigue in workers, excessive moisture indicates water seepage or poor drainage that can weaken the roof and sides of a tunnel, and poor illumination makes it difficult to notice obstacles, roof falls and other hazards. Conventional safety practice relies on periodic inspection by workers carrying portable detectors and on fixed sensing points installed at selected locations.

Both approaches have limitations: inspection puts people in the very places that are most dangerous, and fixed points cannot cover the full length of a changing mine.Advances in low-cost microcontrollers, wireless communication and cloud platforms now make it possible to build small mobile robots that can carry several sensors into hazardous areas and report what they find in real time. The ESP32 is a dual-core microcontroller with built-in Wi-Fi and a 12-bit analog-to-digital converter, which makes it well suited to collecting readings from several analog sensors and to connecting directly to IoT platforms. ThingSpeak provides cloud storage and live graphs of the uploaded data, and Blynk provides a simple mobile interface for controlling the rover. The ESP32-CAM adds low-cost video, so that an operator can see the surroundings as well as the sensor values. These technologies are combined in the proposed rover.

1.2 MOTIVATION

The main motivation of this project is to protect the lives and health of mine workers by removing them from the first line of risk. Manual inspection increases the chance of accidents, poisoning and health problems, and fixed monitoring points cannot give a complete picture of the mine. A mobile rover that continuously measures the environment, provides visual surveillance and raises early warnings is therefore needed to improve mine safety.

The scope of the proposed work is to design and implement a mobile rover that:

- detects toxic and combustible gases and identifies unsafe gas levels;
- monitors temperature, humidity, moisture and light intensity of the surroundings;
- provides real-time visual surveillance using the ESP32-CAM mounted on a servo motor;
- displays the readings locally on an LCD and transmits them to the ThingSpeak cloud;
- is driven remotely in four directions through the Blynk application; and
- reduces direct human exposure to hazardous underground areas.

SUMMARY

Underground coal mining exposes workers to toxic and combustible gases, high temperature, moisture and poor lighting, and manual inspection or fixed monitoring points cannot adequately cover the mine. The proposed IoT-based rover carries gas, temperature, humidity, light and moisture sensors together with a camera, shows the readings on an LCD, uploads them to the ThingSpeak cloud and is driven through the Blynk application.

CHAPTER 2
LITERATURE SURVEY

The literature survey reviews existing research related to IoT-based and robot-assisted monitoring of coal mines, wireless sensor networks for mine safety, long-range communication for underground monitoring, and the general principles of the Internet of Things. The survey helps to understand the monitoring techniques and communication methods already used, to identify their limitations, and to establish the research gap that the proposed rover is intended to fill.

2.1 LITERATURE REVIEW

1. IoT-Based Coal Mine Monitoring Robot

Dudhale et al. developed an ESP32-based robot for coal mine monitoring. The robot measures temperature, humidity, vibration, inclination and hazardous gases, and sends the data to the cloud where alerts can be generated. This work is relevant to the proposed project because it confirms that the ESP32 is a suitable controller for a mobile mine-monitoring robot with cloud connectivity.

2. Hazardous Gas Detection and Environmental Monitoring in Coal Mines Using IoT and GSM-Based Autonomous Robot

Patil et al. proposed an autonomous robot that uses IoT and GSM communication to detect hazardous gases and monitor the environment in coal mines. Emergency alerts are sent when unsafe conditions are detected. This work is relevant to the proposed project because it demonstrates the value of a mobile gas-monitoring robot with automatic alerting.

3. IoT-Based Coal Mine Safety Robot with Casualty Detection and First Aid Kit Supply

Bhargavi et al. presented a safety robot that senses gas, fire and vibration, carries a camera, detects casualties and supplies a first-aid kit. This work is relevant to the proposed project because it shows that camera-based surveillance combined with environmental sensing improves the usefulness of a rescue and inspection robot.

4. Research on Mine Safety Monitoring System Based on WSN

This study applies a wireless sensor network to monitor vibration, temperature, humidity and gas in underground mines. Sensor nodes at fixed positions send their measurements to a central station. This work is relevant to the proposed project because it

identifies the key parameters to be monitored; at the same time, fixed nodes cannot move to unexplored areas, which a rover can.

5. IoT-Based Coal Mine Monitoring System Using LoRa Cloud

Akram developed a coal mine monitoring system that uses LoRa for low-power, long-range data transfer and a cloud platform for data storage and display. This work is relevant to the proposed project because it shows how the ThingSpeak-type cloud platforms can be used for mine data and indicates a possible communication upgrade where Wi-Fi coverage is not available underground.

6. Research Directions for the Internet of Things

Stankovic discussed the major research directions and challenges of the Internet of Things, including scalability, architecture, robustness, security, privacy and the role of humans in the loop. This work is relevant to the proposed project because it provides the general design considerations for connecting sensors and actuators to the cloud reliably and securely.

7. The Internet of Things: A Survey

Atzori, Iera and Morabito surveyed the enabling technologies, middleware and application domains of the Internet of Things. This survey is relevant to the proposed project because it supports the sensor, controller and cloud structure used in the rover and the concept of remote monitoring and control.

2.2 RESEARCH GAP IDENTIFICATION

From the literature survey it is observed that existing systems mostly rely on predefined sensor thresholds and concentrate on a limited set of parameters. Several works are limited to fixed sensing nodes that cannot reach new areas of the mine. Others measure gases and environmental parameters but provide little or no live visual surveillance, so that the operator cannot see the actual condition of the tunnel. Light intensity and moisture, which indicate visibility and water seepage, are rarely monitored together with gases. Many solutions are also built from expensive components or depend on communication infrastructure that is not simple to deploy.

Therefore, an integrated and low-cost mobile system is required that combines toxic-gas detection, environmental monitoring (temperature, humidity, light and moisture), live camera surveillance, local display and cloud logging, and remote driving in a single platform. The proposed rover addresses these gaps by combining MQ-9 and MQ-135 gas sensing, DHT11, LDR and moisture sensing, a servo-mounted ESP32-CAM, ThingSpeak monitoring and Blynk control on an ESP32-based mobile chassis.

SUMMARY

The reviewed works show that IoT, robots and wireless sensor networks can improve mine safety, but most of them rely on fixed thresholds, fixed nodes or limited sensing and offer little visual surveillance. The proposed rover integrates gas, environmental and light sensing with a pan-sweeping camera, local display, cloud monitoring and remote driving, and so fills the identified research gap at low cost.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

This chapter explains the objectives, the overall methodology and the implementation procedure of the proposed work. It describes the major steps involved in developing the underground coal mining rover, including the selection of components, sensing, data display, cloud upload, remote driving and surveillance. It also presents the tools and techniques used for programming, monitoring and performance evaluation.

3.1 OBJECTIVES

The main objective of this work is to develop a low-cost IoT-based mobile rover for toxic gas detection, environmental monitoring and real-time surveillance in underground coal mines, so that the exposure of workers to hazardous areas is reduced and mine safety is improved.

The specific objectives are:

- To design a four-wheeled rover chassis driven by DC geared motors through an L293D motor driver and controlled by an ESP32.
- To detect carbon monoxide, combustible gases (such as methane) and other harmful gases using the MQ-9 and MQ-135 sensors.
- To monitor temperature, humidity, moisture and light intensity using the DHT11, moisture sensor and LDR.
- To provide live visual surveillance using an ESP32-CAM mounted on a servo motor that sweeps from 0° to 180°.
- To display the sensor readings locally on a 16 × 2 LCD and to give an audible alert for hazardous conditions.
- To upload the readings to the ThingSpeak cloud and to drive the rover remotely through the Blynk application.
- To analyse the performance of the prototype through the cloud graphs and hardware output.

3.2 FLOW CHART OF THE PROPOSED WORK

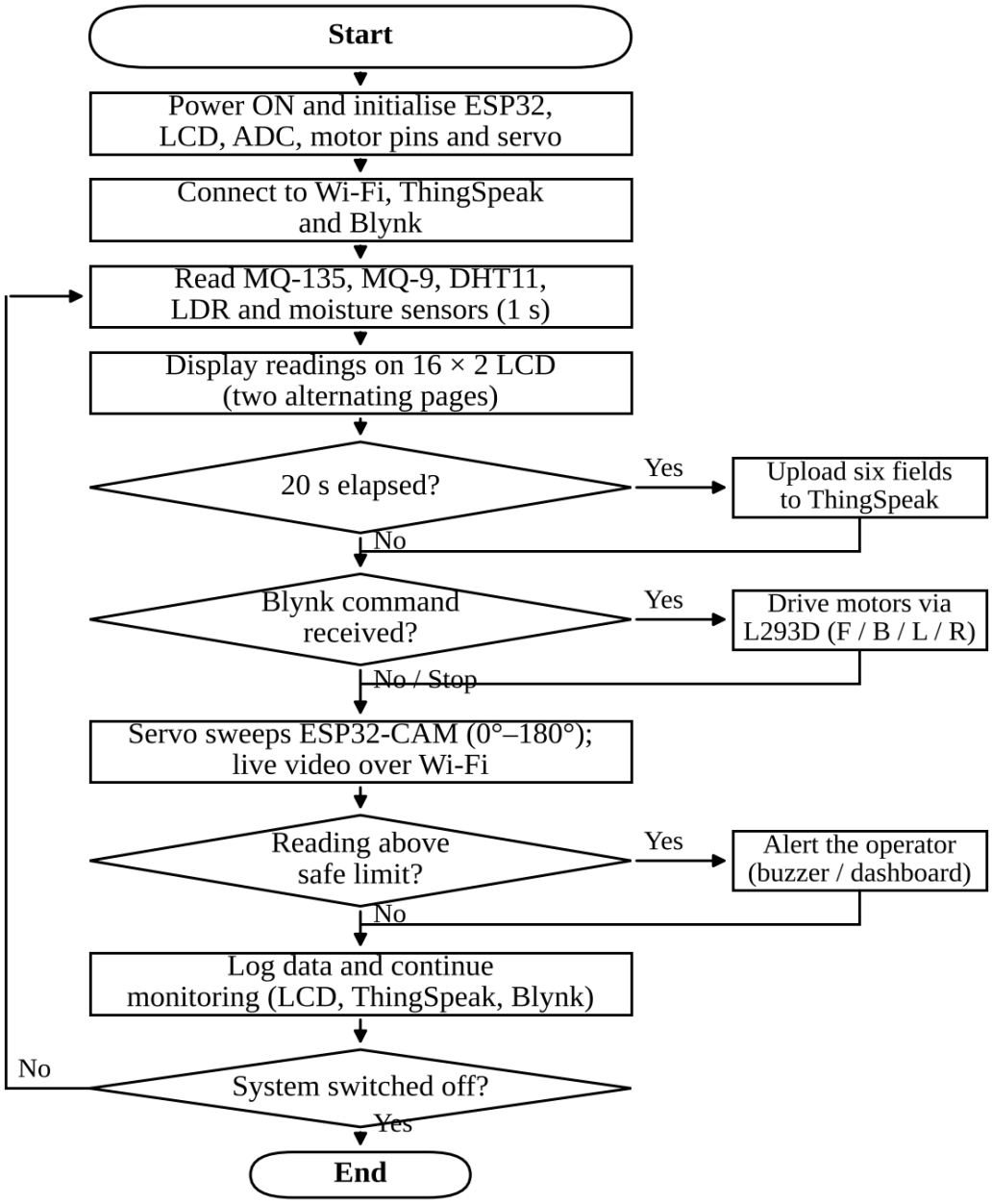


Figure 3.1 Flow Chart of the Proposed Work

3.3 SELECTION OF COMPONENTS, TOOLS, TECHNIQUES, PROCEDURES AND TESTING METHODS

The components and tools used in the proposed system are selected on the basis of their functionality, compatibility, availability, cost and suitability for a battery-powered mobile monitoring platform. Each component performs a specific function in sensing, processing, display, communication, surveillance or motion.

A. Selection of Components

Table 3.1 Selection of Components and their Functions

S.No	Component	Reason for selection / function
1	ESP32 controller	Dual-core controller with built-in Wi-Fi and 12-bit ADC; reads all sensors, drives motors and servo, and communicates with ThingSpeak and Blynk.
2	ESP32-CAM module	Low-cost camera module with Wi-Fi for live video streaming of the underground area.
3	MQ-9 gas sensor	Detects carbon monoxide and combustible gases such as methane; analog output.
4	MQ-135 gas sensor	Detects harmful gases such as ammonia, benzene, smoke and CO_2, indicating air quality; analog output.
5	DHT11 sensor	Measures temperature and relative humidity of the surroundings with a single digital pin.
6	LDR sensor	Measures light intensity to indicate low-visibility conditions.
7	Moisture sensor	Measures the wetness of the surroundings to indicate water seepage.
8	L293D motor driver	H-bridge driver that runs the DC motors in both directions.
9	DC geared motors (12 V, 60 RPM)	Provide the torque and low speed required for the rover to move over rough floor.
10	Servo motor	Sweeps the ESP32-CAM from 0° to 180° for wide-angle surveillance.
11	16 × 2 LCD	Local display of the sensor readings.
12	Buzzer	Audible alert for hazardous conditions.
13	Power supply (transformer, 7812, 7805)	Provides regulated 12 V for the motors and 5 V for the controller and sensors.

B. Software and Tools Selection

- **Arduino IDE** with embedded C/C++ (Arduino framework) is used to write, compile and upload the program to the ESP32.
- **Libraries:** LiquidCrystal for the LCD, DFRobot_DHT11 for the DHT11, WiFi for the network connection, ThingSpeak for cloud upload, BlynkSimpleEsp32 for remote control and ESP32Servo for the servo motor.
- **ThingSpeak** is used for cloud data logging and real-time graphs of the six monitored parameters.
- **Blynk** is used as the mobile application for driving the rover.

C. Data Collection and Monitoring Techniques

The ESP32 reads the four analog sensors through its 12-bit ADC (values 0 to 4095) and the DHT11 through a digital pin, once every second. The LDR and moisture readings are inverted in software (4095 minus the raw value) so that a higher number means more light or more moisture. The readings are printed on the serial monitor, shown on the LCD and, every 20 seconds, written to the ThingSpeak channel as six fields. The 20-second interval is chosen to respect the upload rate of the free ThingSpeak service.

D. Implementation Procedure

1. Assemble the four-wheeled chassis and mount the motors, wheels, driver and power supply.
2. Connect the sensors, LCD, buzzer, servo motor and motor driver to the ESP32 GPIO pins as given in Table 4.1.
3. Mount the ESP32-CAM on the servo motor at the front of the rover.
4. Write and upload the Arduino program that reads the sensors, updates the LCD, sweeps the servo, uploads to ThingSpeak and receives the Blynk commands.
5. Create the ThingSpeak channel with six fields and the Blynk template with four drive buttons (V0 to V3).
6. Power the rover, connect it to Wi-Fi and verify the operation of every function.

E. Testing and Performance Evaluation

The prototype is verified function by function and then as a complete system. The sensor readings are observed on the serial monitor and the LCD, the cloud upload is verified on the ThingSpeak graphs, the drive commands are verified by operating the four Blynk buttons, and the camera sweep is verified on the live video. The response of the gas, temperature, humidity, light and moisture sensors is observed when conditions near the rover change. The ThingSpeak graphs and the hardware output are then analysed to judge the overall performance of the rover.

SUMMARY

This chapter presented the objectives and the methodology of the proposed rover. The flow chart explained the sequence of initialisation, sensing, display, cloud upload, remote driving and surveillance. The components, software tools, data-collection technique, implementation steps and testing methods were described. Together they give a systematic approach to building a low-cost rover for monitoring hazardous underground areas.

CHAPTER 4
PROPOSED WORK MODULES

The proposed system consists of a power supply, gas sensors, environmental sensors, the ESP32 controller, a motor driver with DC motors, a servo-mounted ESP32-CAM, an LCD display and a buzzer, together with the ThingSpeak and Blynk cloud services. These modules are integrated so that the rover can move through a hazardous area, sense its surroundings, show the results locally, send them to the cloud and give the operator a live view.

4.1 PROPOSED WORK

The proposed work focuses on developing an integrated mobile rover for monitoring underground coal mines. A regulated power supply provides 12 V for the motors and 5 V for the electronics. The ESP32 continuously reads the MQ-9, MQ-135, DHT11, LDR and moisture sensors, displays the values on the 16 × 2 LCD and uploads them to the ThingSpeak cloud over Wi-Fi. The rover is driven in four directions from the Blynk application through the L293D motor driver, and the ESP32-CAM, swept by a servo motor between 0° and 180°, streams live video of the surroundings. When a reading exceeds its safe limit, the operator is alerted so that appropriate action can be taken. This integrated approach aims to improve safety, reduce manual inspection and give early warning of hazardous conditions.

4.1.1 Block Diagram of Proposed Work

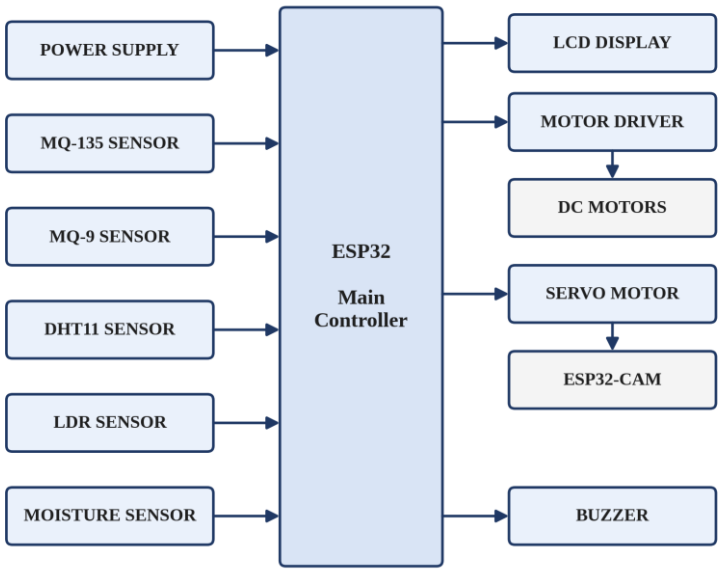


Figure 4.1 Block Diagram of Proposed Work

4.1.2 Block Diagram Description

Figure 4.1 shows the block diagram of the proposed system. The power supply and the sensors feed the ESP32, which controls the LCD, the motor driver, the servo motor and the buzzer. The function of each block is described below.

1. Power Supply

- A step-down transformer, a bridge rectifier and a filter capacitor convert the mains supply into smooth DC voltage.
- The 7812 and 7805 regulators give regulated 12 V and 5 V outputs.
- The 12 V output supplies the DC motors and the 5 V output supplies the controller, sensors and display.

2. Gas Sensors (MQ-9 and MQ-135)

- The MQ-9 detects carbon monoxide and combustible gases such as methane. Its sensing element is heated, and its resistance changes with gas concentration.
- The MQ-135 detects harmful gases such as ammonia, benzene, smoke and CO_2 and is used as an indicator of air quality.
- Both sensors give analog outputs that are read by the ESP32 ADC; a higher value indicates a higher gas level.

3. DHT11, LDR and Moisture Sensors

- The DHT11 measures the temperature and the relative humidity of the surroundings.
- The LDR measures light intensity, and the moisture sensor measures the wetness of the surroundings.
- Abnormal values indicate unsafe, wet or low-visibility working conditions.

4. ESP32 Controller

- It reads all the sensors every second and processes the readings.
- It controls the motors, servo, LCD and buzzer.
- Its Wi-Fi uploads the data to ThingSpeak for graphs and receives Blynk commands for remote control.

5. Motor Driver and DC Motors

- The L293D H-bridge driver controls the DC motors in both directions.
- It allows the rover to move forward, backward, left and right.

6. Servo Motor and ESP32-CAM

- The ESP32-CAM streams live video of the underground area over Wi-Fi.
- The servo motor sweeps the camera between 0° and 180°.
- This gives wider surveillance without moving the rover.

7. LCD Display and Buzzer

- A 16 × 2 LCD displays the sensor readings locally.
- The display alternates between two pages of readings every two seconds.
- The buzzer gives an audible alert for hazardous conditions.

Overall, the proposed system combines sensing, local display, cloud monitoring, remote driving and live surveillance in a single low-cost mobile platform.

4.1.3 Circuit Diagram and Description

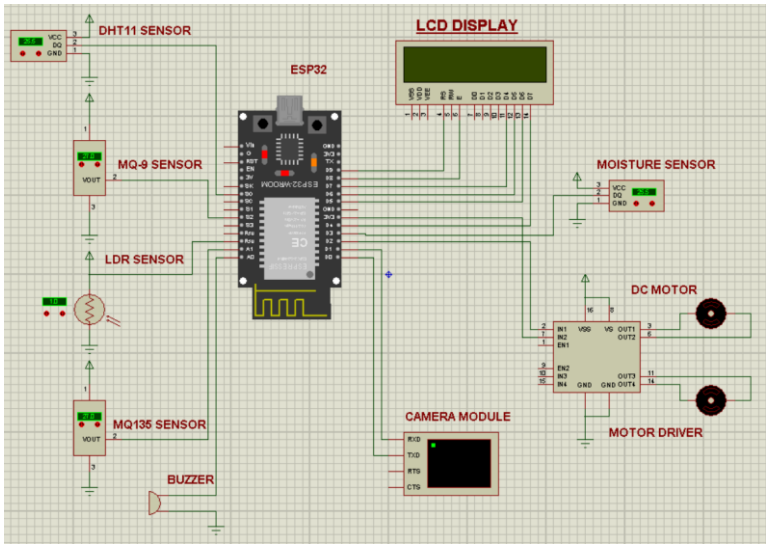


Figure 4.2 Circuit Diagram of Proposed Work

Figure 4.2 illustrates the circuit diagram of the rover, showing the interconnection of the power supply, ESP32, gas, DHT11, LDR and moisture sensors, the L293D motor driver with the DC motors, the servo motor, the LCD display and the buzzer. The sensor outputs are connected to the ADC and digital pins of the ESP32 and the pin assignment is given in Table 4.1. The MQ-135, MQ-9, LDR and moisture sensors use the ADC1 input pins (GPIO 34, 35, 32 and 33), which remain usable while Wi-Fi is active.

Table: 4.1 Pin Connections with the ESP32

Component	Signal	ESP32 GPIO
16 × 2 LCD	RS, E, D4, D5, D6, D7	4, 16, 17, 5, 18, 19
DHT11	Data	13
MQ-135	Analog output	34
MQ-9	Analog output	35
LDR	Analog output	32
Moisture sensor	Analog output	33
L293D motor driver	Inputs IN1 – IN4	14, 27, 26, 25
Servo motor	PWM signal	21

The main electrical characteristics of the sensing and driving devices used in the circuit are listed in Table 4.2.

Table: 4.2 Characteristics of the Main Devices

Device	Parameter	Typical value (datasheet)
ESP32	ADC resolution / supply	12-bit (0 – 4095) / 3.3 V logic
MQ-9	Detects	CO and combustible gases (CH_4, LPG)
MQ-135	Detects	NH_3, benzene, smoke, CO_2
DHT11	Temperature / humidity range	0 – 50 °C (±2 °C) / 20 – 90 % RH (±5 %)
L293D	Output current per channel	600 mA continuous
Servo motor	Control signal	50 Hz PWM, 0° – 180°
ESP32-CAM	Camera	OV2640, 2 MP, Wi-Fi streaming

4.2 METHODOLOGY OF THE PROPOSED WORK

The proposed system follows a systematic methodology that integrates power supply and rover motion, environmental sensing, cloud monitoring, remote control and visual surveillance.

4.2.1 Power Supply and Rover Mobility

The rover obtains its power from a regulated supply made up of a step-down transformer, a bridge rectifier, a filter and the 7812 and 7805 regulators. The 12 V output drives the geared DC motors and the 5 V output feeds the ESP32, the sensors and the display.

The four-wheeled chassis is driven by two geared motors (12 V, 60 RPM) through the L293D H-bridge driver. The ESP32 sets the four driver inputs (GPIO 14, 27, 26 and 25) to move the rover forward, backward, left or right. For forward and backward motion both sides turn in the same direction, and for turning the two sides turn in opposite directions so that the rover turns about its own axis. This is useful in narrow tunnels where there is little room to manoeuvre. The slow 60 RPM motors give high torque and smooth, controllable movement.

4.2.2 Parameter Monitoring and Hazard Detection

Continuous monitoring is essential for identifying abnormal conditions in an underground mine. The ESP32 reads the MQ-135 and MQ-9 gas sensors, the LDR and the moisture sensor through its ADC and the DHT11 through a digital pin once every second. The gas sensors indicate harmful and combustible gas levels, the DHT11 gives temperature and humidity, the LDR indicates how bright or dark the area is, and the moisture sensor indicates water seepage. A rise in the gas readings, a high temperature or humidity, or a high moisture value is treated as a sign of an unsafe condition, and the operator is alerted when a reading exceeds its safe limit. The same values are also available as graphs on ThingSpeak, so that trends can be studied and the safe limits for each sensor can be set from observed readings.

4.2.3 Cloud Monitoring and Remote Control

The ESP32 connects to the Wi-Fi network and uploads six fields (temperature, humidity, harmful gases, carbon monoxide/combustible gases, light and moisture) to the ThingSpeak channel every 20 seconds. ThingSpeak stores the data and plots it as real-time graphs that can be viewed from anywhere. For remote control, the Blynk application sends commands to the ESP32 through four virtual pins (V0 to V3), one each for forward, reverse, left and right. When a button is pressed, the corresponding motor pattern is applied; when it is released, the stopRobot() function switches all motor outputs off, so the rover stops immediately.

4.2.4 Surveillance, Hardware Implementation and Experimental Setup

The ESP32-CAM is mounted at the front of the rover on a servo motor that sweeps from 0° to 180° in 1° steps every 40 ms. This gives a wide field of view without moving the rover, and the live video is streamed over Wi-Fi to the operator. The sensors, LCD, buzzer and motor driver are fitted on the chassis and connected to the ESP32 GPIO pins, and the power supply provides 12 V and 5 V. The assembled prototype is shown in Figure 4.3.

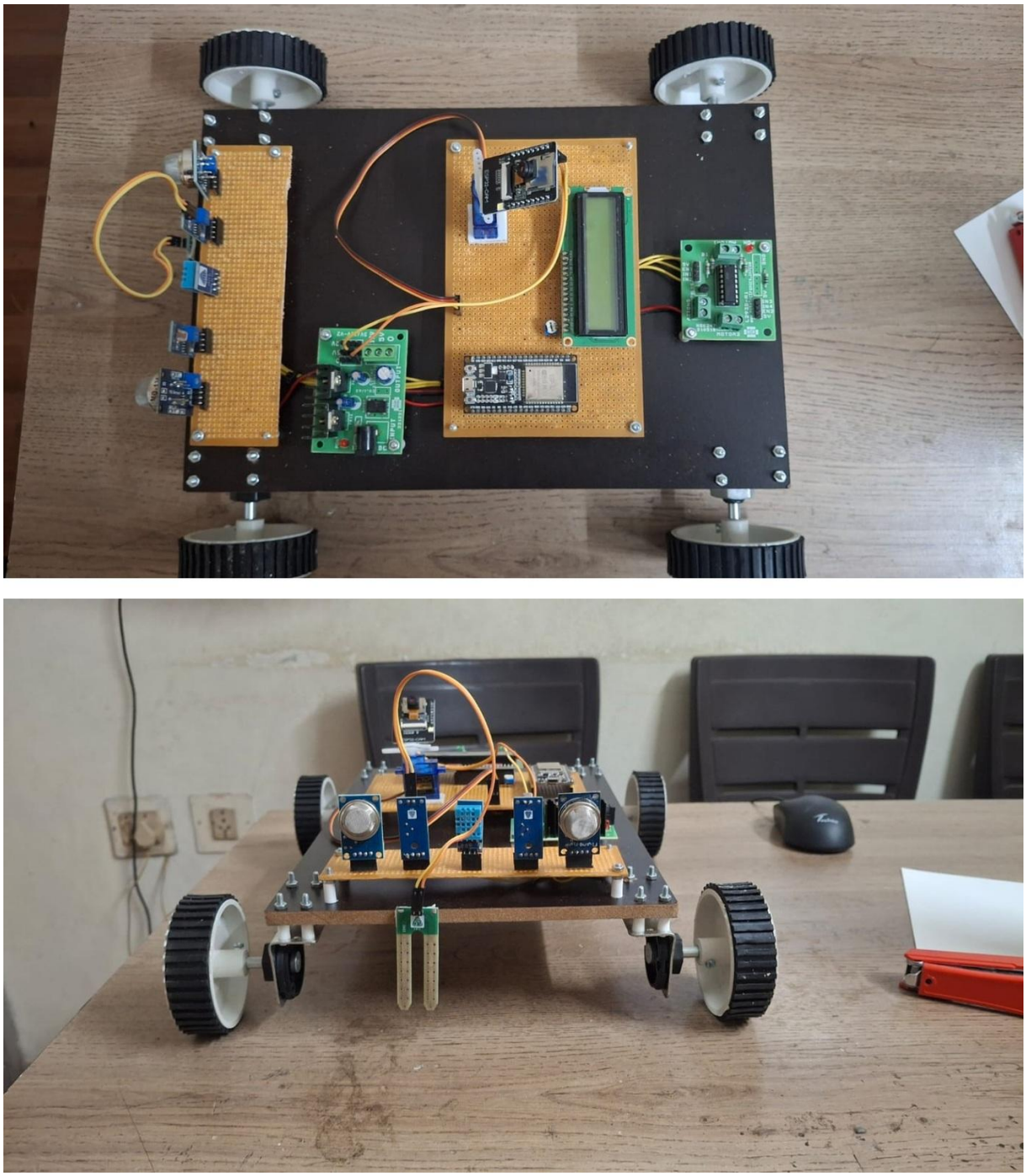


Figure: 4.3 Experimental Setup

When the rover is switched on, the LCD shows a start message and the Wi-Fi connection status. The ESP32 then reads the sensors every second, alternates the LCD between two pages every two seconds, uploads to ThingSpeak every 20 seconds and responds to the Blynk commands while the servo sweeps the camera. When a reading exceeds its safe limit, the user is alerted about the hazardous condition.

4.3 SOFTWARE IMPLEMENTATION

The software is written in embedded C/C++ in the Arduino IDE and uploaded to the ESP32. The program is organised into small functions that are called from the main loop according to timers, so that sensing, display, upload and servo movement can all run together without blocking the Blynk connection. The complete listing is given in the Appendix.

4.3.1 Software Flow Chart

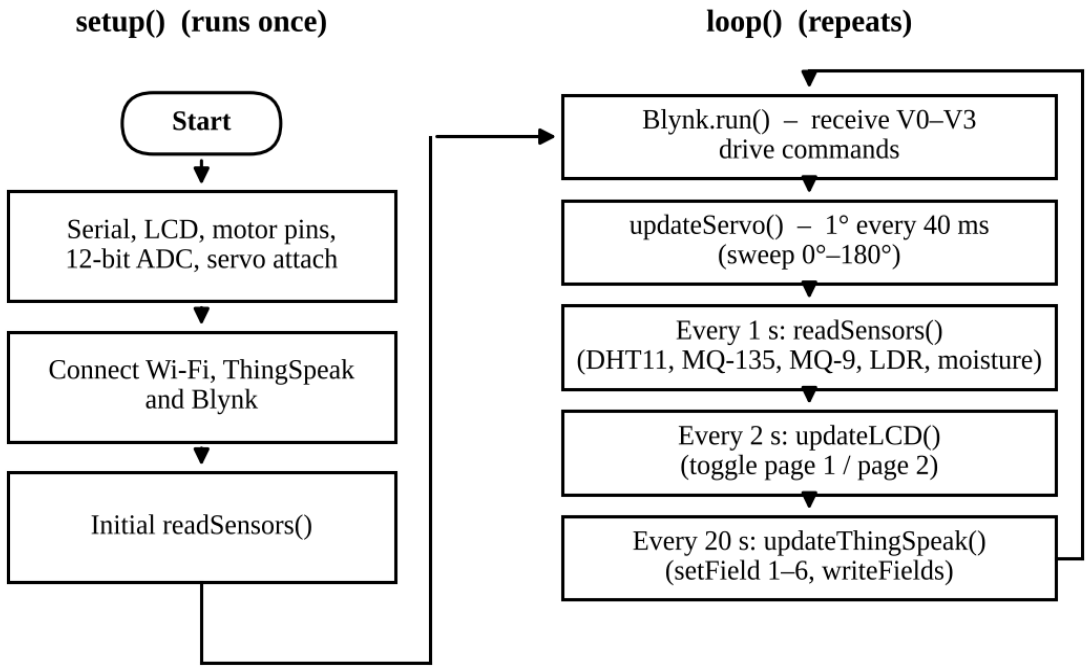


Figure 4.4 Software Flow Chart

4.3.2 Software Description

The program starts with the setup() function, which opens the serial port, initialises the LCD, sets the four motor pins as outputs and stops the rover, sets the ADC to 12-bit resolution and attaches the servo (50 Hz, 500–2400 µs pulses). It then shows a start message, connects to the Wi-Fi network, starts ThingSpeak, configures Blynk and takes an initial sensor reading.The loop() function then repeats the following tasks.

The four Blynk virtual pins are mapped to the motor outputs as shown in Table 4.3. When the button value is 1 the pattern is applied, and any other value calls stopRobot().

Table: 4.3 Blynk Virtual Pins and Motor Output Patterns

Blynk pin	Action	Motor 1 (GPIO 14)	Motor 2 (GPIO 27)	Motor 3 (GPIO 26)	Motor 4 (GPIO 25)
V0	Forward	HIGH	LOW	LOW	HIGH
V1	Reverse	LOW	HIGH	HIGH	LOW
V2	Left	LOW	HIGH	LOW	HIGH
V3	Right	HIGH	LOW	HIGH	LOW

SUMMARY

The proposed rover is made up of a regulated power supply, gas and environmental sensors, an ESP32 controller, an L293D-driven chassis, a servo-mounted ESP32-CAM, an LCD and a buzzer. The ESP32 reads the sensors every second, updates the LCD, uploads the data to ThingSpeak every 20 seconds and moves the rover according to Blynk commands, while the servo sweeps the camera for live surveillance. The software is organised into timer-based functions so that all tasks run together.

CHAPTER 5
RESULTS AND DISCUSSION

This chapter presents the results obtained from the hardware prototype of the rover. The parameters recorded on the ThingSpeak channel, the hardware output and the functional performance of the rover are discussed, followed by the significance, strengths and limitations of the work and the cost analysis.

5.1 RESULTS

The ThingSpeak channel records temperature, humidity, harmful gas level, carbon monoxide/combustible gas level, light intensity and moisture as six separate fields (Table 5.1) and displays them as real-time graphs.

Table: 5.1 Parameters Monitored and ThingSpeak Fields

Field	Parameter	Sensor	Unit
Field 1	Temperature	DHT11	°C
Field 2	Humidity	DHT11	% RH
Field 3	Harmful gases / air quality	MQ-135	ADC value
Field 4	Carbon monoxide / combustible gases	MQ-9	ADC value
Field 5	Light intensity	LDR	ADC value
Field 6	Moisture level	Moisture sensor	ADC value

5.1.1 Software Output (ThingSpeak Graphs / Blynk Dashboard)



Figure 5.1 ThingSpeak Field 1 – Temperature monitoring

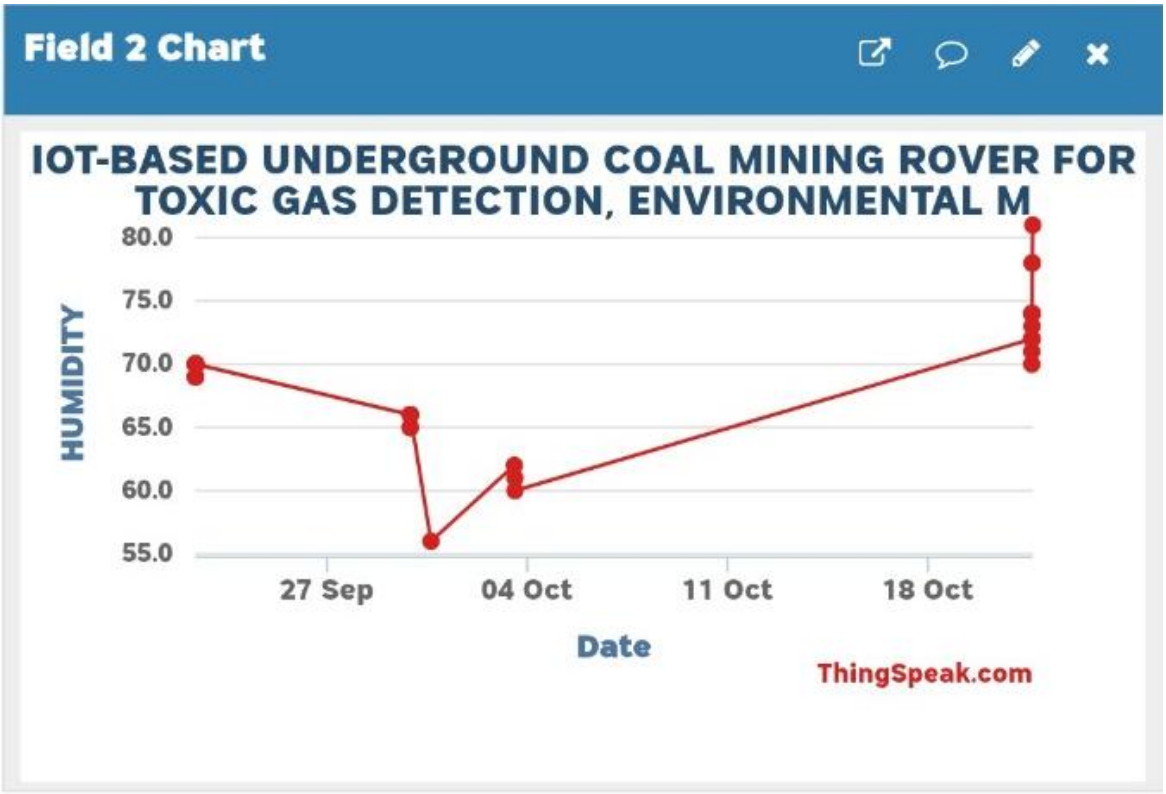


Figure 5.2 ThingSpeak Field 2 – Humidity monitoring



Figure 5.3 ThingSpeak Field 3 – Air-quality monitoring (MQ-135)



Figure 5.4 ThingSpeak Field 4 – MQ-9 gas-sensor monitoring

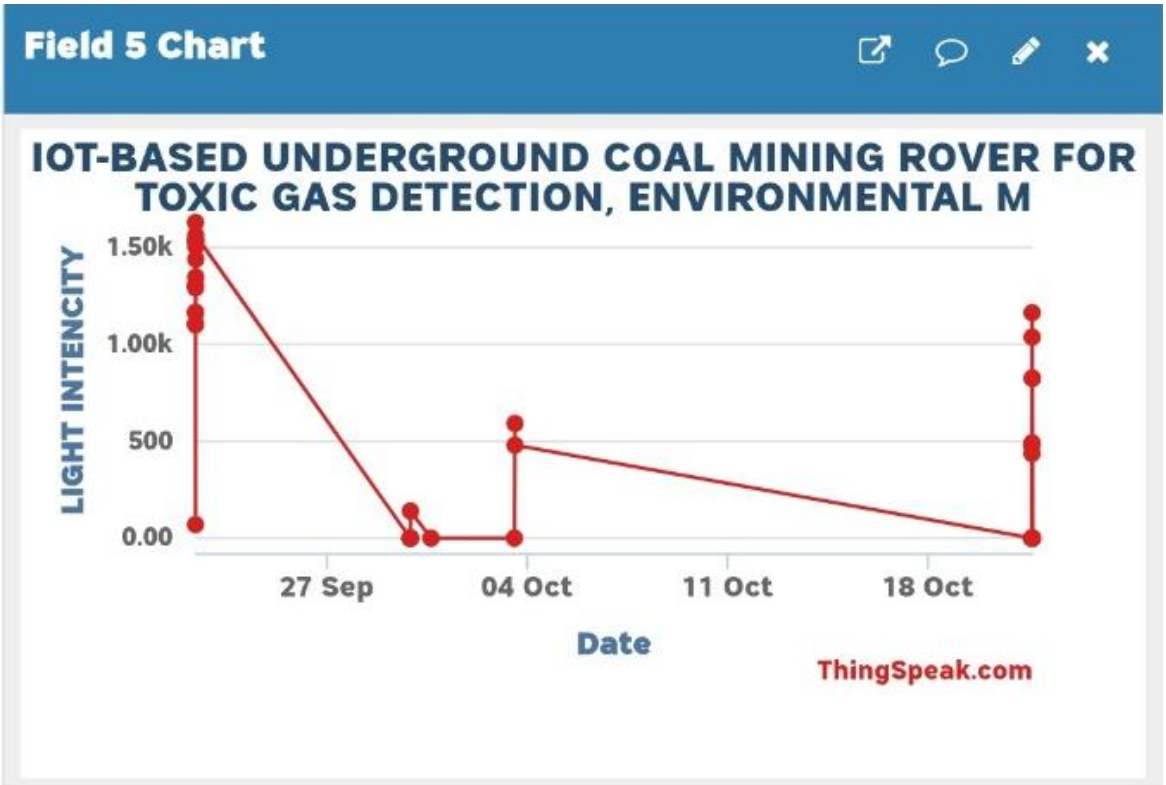


Figure 5.5 ThingSpeak Field 5 – Light-intensity monitoring (LDR)



Figure 5.6 ThingSpeak Field 6 – Moisture-level monitoring

Blynk Application Output

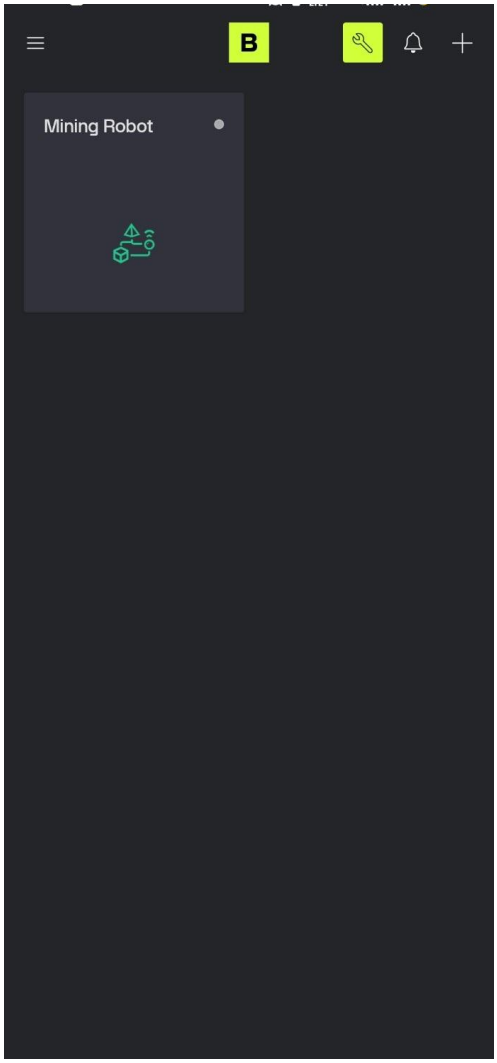


Figure 5.7 Blynk dashboard for the mining rover

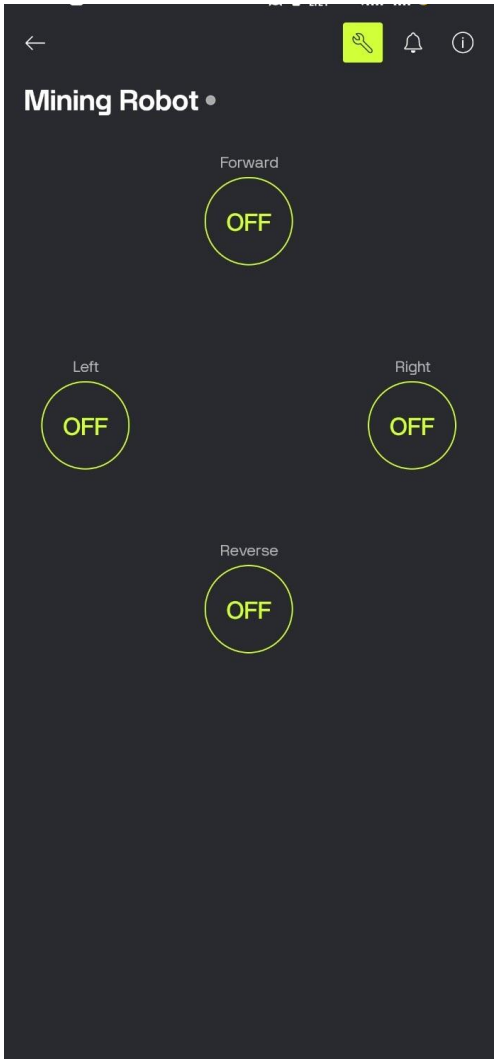


Figure 5.8 Blynk remote movement-control interface

Figures 5.1–5.6 show the six ThingSpeak field charts for temperature, humidity, air quality, MQ-9 gas response, light intensity and moisture level. Figures 5.7 and 5.8 show the Blynk dashboard and the remote movement-control interface used to operate the rover.

5.1.2 Hardware Output

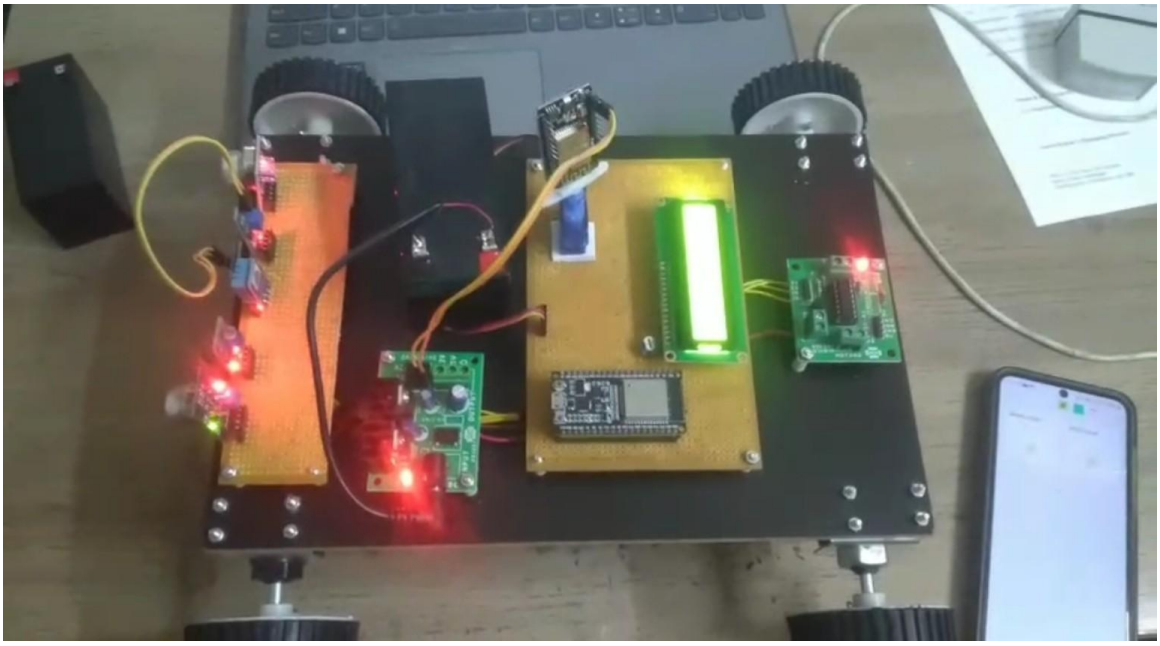


Figure 5.9 Hardware Output in Working Condition

Figure 5.9 shows the output of the hardware prototype. The rover successfully measures the gas, temperature, humidity, light and moisture values, displays them on the LCD in two alternating pages, uploads them to the cloud and provides live video for remote surveillance. The LCD makes the readings available at the rover itself, while the cloud graphs make them available to a remote operator.

5.1.3 Functional Verification

The functions of the prototype were verified one by one as summarised in Table 5.2.

Table: 5.2 Functional Verification of the Rover

Function	Method of checking	Observation
Gas sensing	MQ-135 and MQ-9 readings on LCD and serial monitor	Values measured and displayed
Temperature and humidity	DHT11 reading on LCD and ThingSpeak	Values measured and displayed
Light and moisture	LDR and moisture readings on LCD and ThingSpeak	Values measured and displayed
Local display	16 × 2 LCD pages every 2 s	Two pages alternate correctly
Cloud upload	ThingSpeak graphs (every 20 s)	Six fields uploaded and plotted
Remote driving	Blynk buttons V0 – V3	Rover moves in four directions and stops on release
Surveillance	Servo sweep with ESP32-CAM video	Live video of the surroundings obtained

Table: 5.3 Comparison of Existing and Proposed Systems

S.No	Parameter	Existing systems	Proposed rover
1	Mobility	Fixed sensing nodes or manual inspection	Mobile four-wheeled rover driven remotely
2	Gas detection	Limited gas sensing	MQ-9 (CO, combustible gases) and MQ-135 (harmful gases)
3	Environmental sensing	Mostly temperature and humidity	Temperature, humidity, light and moisture
4	Visual surveillance	Absent or limited	Servo-swept ESP32-CAM with live video
5	Monitoring	Local indication or alarm	LCD plus ThingSpeak cloud graphs
6	Remote control	Not available in fixed nodes	Blynk application (forward, reverse, left, right)
7	Human exposure	Workers enter the hazardous area	Workers stay at a safe distance
8	Cost	Commercial robots are expensive	Low-cost prototype (about Rs. 3810)

5.2 SIGNIFICANCE, STRENGTHS AND LIMITATIONS OF THE PROPOSED WORK

Significance

The proposed rover shows that a low-cost IoT platform can take over a dangerous task from human workers. By collecting gas, environmental and visual information from inside a mine and presenting it remotely, it supports early identification of hazards, quicker decisions and safer inspection. The recorded data can also be used later to study the conditions of different sections of the mine.

Strengths

- Reduces the exposure of workers to toxic gases and unsafe areas.
- Combines gas, temperature, humidity, light and moisture sensing in one rover.
- Provides live visual surveillance with a wide sweep using a servo-mounted camera.
- Shows readings locally on the LCD and remotely on ThingSpeak graphs.
- Remote driving through the Blynk mobile application.
- Low cost, built from readily available components, and easy to extend.

Limitations

- The Wi-Fi link used for ThingSpeak, Blynk and video may not be available deep inside a mine.
- MQ-series sensors give analog values that depend on temperature and humidity and need calibration and warm-up before accurate gas concentrations can be read.
- The DHT11 has limited accuracy and range compared with industrial temperature sensors.
- The prototype is not certified as intrinsically safe, and its electronics could themselves ignite methane in a real mine unless suitably enclosed.
- The rover is driven manually and does not yet avoid obstacles automatically.
- The free ThingSpeak service limits how often data can be uploaded.

5.3 COST BENEFIT ANALYSIS

The cost of the components used to build the prototype is given in Table 5.4. The total cost of the prototype is approximately Rs. 3810.

Table: 5.4 Cost Benefit Analysis

S.No	Components Used	Qty	Cost (in Rs.)
1	ESP32 Controller	1	450
2	ESP32-CAM Module	1	700
3	MQ-9 Gas Sensor	1	200
4	MQ-135 Gas Sensor	1	150
5	DHT11 Temperature and Humidity Sensor	1	100
6	LDR Sensor	1	20
7	Moisture Sensor	1	80
8	L293D Motor Driver	1	100
9	DC Geared Motors (12 V, 60 RPM)	2	700
10	Servo Motor	1	150
11	16 × 2 LCD Display	1	180
12	Buzzer	1	30
13	Power Supply (Transformer, 7812, 7805)	1	300
14	Chassis and Wheels	1	500
15	Connecting Wires and Prototype Board	1	150
	TOTAL (APPROXIMATE)		**3810**

The benefit of the rover is far greater than its cost. A commercial inspection robot with comparable sensing and a camera is expensive, whereas this prototype is affordable even for small mines. The main benefit is the safety of workers: every inspection that the rover carries out removes a person from a hazardous tunnel. Early detection of dangerous gas levels can prevent poisoning, fires and explosions, which would otherwise cause loss of life, damage to equipment and long stoppages of production.

SUMMARY

The hardware prototype measures the gas, temperature, humidity, light and moisture values, displays them on the LCD, uploads them to ThingSpeak as six fields, moves in four directions under Blynk control and gives live video through the servo-mounted ESP32-CAM. The rover costs about Rs. 3810. Its limitations, chiefly Wi-Fi coverage, sensor calibration and manual driving, point to the improvements listed in the next chapter.

CHAPTER 6
CONCLUSION AND SUGGESTIONS FOR FUTURE WORK

The IoT-Based Underground Coal Mining Rover integrates gas detection, environmental monitoring, live surveillance, cloud logging and remote driving in a low-cost mobile platform. This chapter gives the conclusion drawn from the work and suggests improvements for the future.

6.1 CONCLUSION

The IoT-Based Underground Coal Mining Rover provides a practical approach for remote monitoring of underground coal mines. It combines MQ-9 and MQ-135 gas sensing with temperature, humidity, light and moisture sensing, ESP32-CAM surveillance and ThingSpeak cloud monitoring, and it is driven remotely through the Blynk application. The ESP32 reads the sensors every second, shows them on a 16 × 2 LCD, uploads six fields to the cloud every 20 seconds and moves the rover according to the commands received. The prototype, built for about Rs. 3810, helps identify hazardous conditions at an early stage and reduces the need for workers to enter unsafe areas. Overall, the system provides an integrated and affordable approach to improving safety in underground coal mines.

6.2 SUGGESTIONS FOR FUTURE WORK

The proposed system can be further improved by adding the following features:

1. **AI Integration:** Predict hazardous conditions from the stored sensor data and give early warnings before limits are reached.
2. **Advanced Sensors:** Add more accurate and calibrated methane and carbon monoxide sensors and use an industrial temperature sensor.
3. **Long-Range Communication:** Use LoRa or a mesh network where Wi-Fi is not available underground.
4. **Autonomous Navigation:** Add ultrasonic or LiDAR sensors for obstacle detection and automatic path following.
5. **Mobile Application:** Provide live data, video and emergency notifications on a phone, with automatic alerts when safe limits are crossed.
6. **Safe Enclosure:** House the electronics in a flameproof, dust-proof and water-resistant enclosure suitable for use in a methane atmosphere.

7. **Solar-Assisted Power:** Use a battery with solar charging for longer operation at the mine entrance and surface stations.
8. **Field Testing:** Test the rover under real mine conditions and calibrate the safe limits for each sensor.

SUMMARY

The IoT-Based Underground Coal Mining Rover combines gas, temperature, humidity, light and moisture sensing with ESP32-CAM surveillance and ThingSpeak monitoring, helping to identify hazardous conditions early and reducing the need for workers to enter unsafe areas. Future work includes AI-based prediction, more accurate sensors, LoRa or mesh communication, autonomous navigation, a mobile application with emergency notifications, a safe enclosure and field testing.

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APPENDIX: PROGRAM CODE

The complete Arduino program of the rover is given below. The Wi-Fi credentials, Blynk token and ThingSpeak API key are replaced by placeholders.

```
#define BLYNK_TEMPLATE_ID "YOUR_BLYNK_TEMPLATE_ID"
#define BLYNK_TEMPLATE_NAME "MINING ROBOT"
#define BLYNK_AUTH_TOKEN "YOUR_BLYNK_AUTH_TOKEN"
#include <LiquidCrystal.h>
#include <DFRobot_DHT11.h>
#include <WiFi.h>
#include <ThingSpeak.h>
#include <BlynkSimpleEsp32.h>
#include <ESP32Servo.h>

// ---------- LCD ----------
LiquidCrystal lcd(4, 16, 17, 5, 18, 19);

// ---------- DHT11 ----------
DFRobot_DHT11 DHT;
#define DHT11_PIN 13

// ---------- SENSORS ----------
#define MQ135_PIN 34
#define MQ9_PIN 35
#define LDR_PIN 32
#define MOISTURE_PIN 33
int t1 = 0;
int h1 = 0;
int mq135_value = 0;
int mq9_value = 0;
int ldr_value = 0;
int moisture_value = 0;

// ---------- MOTORS ----------
```

```
int motor1 = 14;
int motor2 = 27;
int motor3 = 26;
int motor4 = 25;

// ---------- SERVO ----------
// Servo signal connected to GPIO 21
#define SERVO_PIN 21
Servo scanServo;
int servoAngle = 0;
int servoDirection = 1;
unsigned long lastServoMove = 0;
// Higher value = slower servo movement
const unsigned long servoInterval = 40;

// ---------- WIFI ----------
const char* WIFI_SSID = "YOUR_WIFI_SSID";
const char* WIFI_PASSWORD = "YOUR_WIFI_PASSWORD";

// ---------- THINGSPEAK ----------
unsigned long channelID = YOUR_CHANNEL_ID;
const char* THINGSPEAK_API_KEY = "YOUR_THINGSPEAK_WRITE_API_KEY";
WiFiClient client;

// ---------- THINGSPEAK TIMER ----------
unsigned long lastThingSpeak = 0;
const unsigned long thingSpeakInterval = 20000;

// ---------- SENSOR TIMER ----------
unsigned long lastSensorRead = 0;
const unsigned long sensorInterval = 1000;

// ---------- LCD TIMER ----------
unsigned long lastLCD = 0;
```

```
const unsigned long lcdInterval = 2000;
int lcdPage = 0;

// ---------- STOP ROBOT ----------
void stopRobot()
{
  digitalWrite(motor1, LOW);
  digitalWrite(motor2, LOW);
  digitalWrite(motor3, LOW);
  digitalWrite(motor4, LOW);
  Serial.println("ROBOT STOP");
}

// ---------- FORWARD - BLYNK V0 ----------
BLYNK_WRITE(V0)
{
  int value = param.asInt();
  if (value == 1)
  {
    digitalWrite(motor1, HIGH);
    digitalWrite(motor2, LOW);
    digitalWrite(motor3, LOW);
    digitalWrite(motor4, HIGH);
    Serial.println("FORWARD");
  }
  else
  {
    stopRobot();
  }
}

// ---------- REVERSE - BLYNK V1 ----------
BLYNK_WRITE(V1)
{
```

```
  int value = param.asInt();
  if (value == 1)
  {
    digitalWrite(motor1, LOW);
    digitalWrite(motor2, HIGH);
    digitalWrite(motor3, HIGH);
    digitalWrite(motor4, LOW);
    Serial.println("REVERSE");
  }
  else
  {
    stopRobot();
  }
}

// ---------- LEFT - BLYNK V2 ----------
BLYNK_WRITE(V2)
{
  int value = param.asInt();
  if (value == 1)
  {
    // Left side backward
    digitalWrite(motor1, LOW);
    digitalWrite(motor2, HIGH);
    // Right side forward
    digitalWrite(motor3, LOW);
    digitalWrite(motor4, HIGH);
    Serial.println("LEFT");
  }
  else
  {
    stopRobot();
  }
}
```

```
// ---------- RIGHT - BLYNK V3 ----------
BLYNK_WRITE(V3)
{
  int value = param.asInt();
  if (value == 1)
  {
    // Left side forward
    digitalWrite(motor1, HIGH);
    digitalWrite(motor2, LOW);
    // Right side backward
    digitalWrite(motor3, HIGH);
    digitalWrite(motor4, LOW);
    Serial.println("RIGHT");
  }
  else
  {
    stopRobot();
  }
}

// ---------- UPDATE SERVO ----------
void updateServo()
{
  if (millis() - lastServoMove >= servoInterval)
  {
    lastServoMove = millis();
    servoAngle += servoDirection;
    // Reach 180 degrees
    if (servoAngle >= 180)
    {
      servoAngle = 180;
      servoDirection = -1;
    }
```

```
        // Reach 0 degrees
        if (servoAngle <= 0)
        {
            servoAngle = 0;
            servoDirection = 1;
        }
        scanServo.write(servoAngle);
    }
}

// ---------- READ SENSORS ----------
void readSensors()
{
    // DHT11
    DHT.read(DHT11_PIN);
    t1 = DHT.temperature;
    h1 = DHT.humidity;
    // MQ135
    mq135_value = analogRead(MQ135_PIN);
    // MQ9
    mq9_value = analogRead(MQ9_PIN);
    // LDR
    // More light = higher value
    ldr_value = 4095 - analogRead(LDR_PIN);
    // Moisture
    // More moisture = higher value
    moisture_value = 4095 - analogRead(MOISTURE_PIN);
    // ---------- SERIAL MONITOR ----------
    Serial.println("----------------------------");
    Serial.print("Temperature: ");
    Serial.print(t1);
    Serial.println(" C");
    Serial.print("Humidity: ");
    Serial.print(h1);
```

```
    Serial.println(" %");
    Serial.print("MQ135: ");
    Serial.println(mq135_value);
    Serial.print("MQ9: ");
    Serial.println(mq9_value);
    Serial.print("LDR: ");
    Serial.println(ldr_value);
    Serial.print("Moisture: ");
    Serial.println(moisture_value);
    Serial.print("Servo Angle: ");
    Serial.println(servoAngle);
}

// ---------- UPDATE LCD ----------
void updateLCD()
{
    if (millis() - lastLCD >= lcdInterval)
    {
        lastLCD = millis();
        lcd.clear();
        // ---------- PAGE 1 ----------
        if (lcdPage == 0)
        {
            lcd.setCursor(0, 0);
            lcd.print("T:");
            lcd.print(t1);
            lcd.print("C H:");
            lcd.print(h1);
            lcd.print("%");
            lcd.setCursor(0, 1);
            lcd.print("G1:");
            lcd.print(mq135_value);
            lcd.print(" G2:");
            lcd.print(mq9_value);
```

```
            lcdPage = 1;
        }
        // ---------- PAGE 2 ----------
        else
        {
            lcd.setCursor(0, 0);
            lcd.print("LDR:");
            lcd.print(ldr_value);
            lcd.setCursor(0, 1);
            lcd.print("Moist:");
            lcd.print(moisture_value);
            lcdPage = 0;
        }
    }
}

// ---------- UPDATE THINGSPEAK ----------
void updateThingSpeak()
{
    if (millis() - lastThingSpeak >= thingSpeakInterval)
    {
        lastThingSpeak = millis();
        if (WiFi.status() == WL_CONNECTED)
        {
            // Field 1 - Temperature
            ThingSpeak.setField(1, t1);
            // Field 2 - Humidity
            ThingSpeak.setField(2, h1);
            // Field 3 - MQ135
            ThingSpeak.setField(3, mq135_value);
            // Field 4 - MQ9
            ThingSpeak.setField(4, mq9_value);
            // Field 5 - LDR
            ThingSpeak.setField(5, ldr_value);
```

```
            // Field 6 - Moisture
            ThingSpeak.setField(6, moisture_value);
            // Send data
            int response = ThingSpeak.writeFields(
            channelID,
            THINGSPEAK_API_KEY
            );
            if (response == 200)
            {
               Serial.println("ThingSpeak Update Successful");
            }
            else
            {
               Serial.print("ThingSpeak Error: ");
               Serial.println(response);
            }
         }
         else
         {
            Serial.println("WiFi Disconnected");
         }
      }
}

// ---------- SETUP ----------
void setup()
{
   // ---------- SERIAL ----------
   Serial.begin(115200);
   // ---------- LCD ----------
   lcd.begin(16, 2);
   // ---------- MOTORS ----------
   pinMode(motor1, OUTPUT);
   pinMode(motor2, OUTPUT);
```

```
pinMode(motor3, OUTPUT);
pinMode(motor4, OUTPUT);
stopRobot();
// ---------- ADC ----------
analogReadResolution(12);
// ---------- SERVO ----------
scanServo.setPeriodHertz(50);
scanServo.attach(
SERVO_PIN,
500,
2400
);
servoAngle = 0;
servoDirection = 1;
scanServo.write(0);
// ---------- STARTING MESSAGE ----------
lcd.setCursor(0, 0);
lcd.print("Coal Mining");
lcd.setCursor(0, 1);
lcd.print("Robot Starting");
delay(2000);
lcd.clear();
// ---------- WIFI ----------
lcd.setCursor(0, 0);
lcd.print("Connecting WiFi");
WiFi.begin(
WIFI_SSID,
WIFI_PASSWORD
);
while (WiFi.status() != WL_CONNECTED)
{
   delay(500);
   Serial.print(".");
}
```

```
    Serial.println();
    Serial.println("WiFi Connected");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("WiFi Connected");
    delay(1500);
    lcd.clear();
    // ---------- THINGSPEAK ----------
    ThingSpeak.begin(client);
    // ---------- BLYNK ----------
    Blynk.config(BLYNK_AUTH_TOKEN);
    if (Blynk.connect())
    {
       Serial.println("Blynk Connected");
    }
    else
    {
       Serial.println("Blynk Connection Failed");
    }
    // ---------- INITIAL SENSOR READ ----------
    readSensors();
}

// ---------- LOOP ----------
void loop()
{
    // Keep Blynk running
    Blynk.run();
    // Servo continuously sweeps
    updateServo();
    // Read sensors every 1 second
    if (millis() - lastSensorRead >= sensorInterval)
    {
       lastSensorRead = millis();
```

```
        readSensors();
    }
    // LCD pages
    updateLCD();
    // ThingSpeak every 20 seconds
    updateThingSpeak();
}
```